Hartejsingh Sandhu

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EXPERIENCE

Committee Member · Developer Club

- Architected and maintained the club's event registration portal using React and Node.js, streamlining attendee data management for 200+ participants.

PROJECTS

HireAI · Team Project · GitHub · resume-screening-platform.vercel.app

- Engineered an AI-driven resume screening tool using Python and PyTorch, automating the extraction of key candidate skills and experience.
- Leveraged AI capabilities to automate the parsing and evaluation of applicant resumes.

NST Helper · Node.js, Express.js, React.js · GitHub · nsthelper.vercel.app

- Engineered a full-stack academic productivity platform for first-year students, centralizing assignment tracking and resource management to improve study efficiency.
- Implemented a scalable Node.js and Express.js backend to handle concurrent user requests, ensuring 99% data retrieval accuracy for assignment tracking.
- Designed an intuitive React.js interface, integrating AI-driven features that reduced manual search time for academic resources by 25%.

URL Shortener · Node.js, Express.js, React.js, MongoDB, Redis · GitHub · urlshortnerhartej.vercel.app

- Built a full-stack URL shortening service to convert long URLs into easily shareable links.
- Implemented Redis caching to reduce database query latency by 40%, significantly improving redirection response times under high load.
- Used MongoDB for persistent storage of original and shortened URL mappings.

Tokenizer · Python · GitHub

- Engineered three custom tokenizers in Python, including a regex-based GPT-2 implementation and a GPT-4-style tokenizer using tiktoken to optimize text processing pipelines.
- Implemented Byte-Pair Encoding (BPE) from scratch to analyze LLM tokenization efficiency, reducing vocabulary overhead and improving text processing speed by 15%.

Distil-BERT · Python, PyTorch · GitHub

- Implemented DistilBERT from scratch through knowledge distillation, importing a pretrained BERT teacher model from Hugging Face and building a custom lightweight student transformer architecture.
- Built custom transformer layers and a dedicated loss module to combine soft-label distillation loss with the student model's task loss during training.

SKILLS

Programming Languages: Python, JavaScript

Developer Tools: Git, GitHub, MongoDB, PostgreSQL, Redis, PyTorch, Vercel

Libraries / Frameworks: React.js, HTML, CSS, Node.js, Express.js, Hugging Face Transformers, tiktoken

Core Competencies: Competitive Programming, Data Structures & Algorithms, REST APIs, NLP, Deep Learning

EDUCATION

Newton School of Technology, Pune **2029**

B.Tech, Computer Science

- CGPA: 9.41 / 10
- SGPA: 9.68 / 10